

Architecture Flow

The architecture follows an identity-driven privileged access model. Enterprise users authenticate through Okta with MFA and are federated into AWS through AWS IAM Identity Center. The assigned permission set maps the user to an AWS IAM role with temporary credentials.

When privileged access is required, the operator submits an access request through the approved ITSM workflow. After approval, the PAM control layer governs privileged access using JIT, credential, session, and policy controls.

HashiCorp Vault provides centralized secrets management, policy enforcement, credential lifecycle control, and audit capability. Access to protected workloads follows a controlled administrative path through a gateway or bastion rather than direct unrestricted access.

AWS CloudTrail and security logging provide an independent audit trail, allowing identity, role, action, and time to be correlated for validation and investigation.

The design demonstrates separation of identity, authentication, authorization, privilege management, secrets management, access control, and audit functions in an enterprise-style security architecture.

IAM: Okta + AWS IAM Identity Center authenticate users, federate access, and enforce role-based authorization with temporary credentials.

PAM: Approved privileged requests flow through JIT/credential/session controls, with HashiCorp Vault managing protected secrets and privileged access.

IAM answers: “Who are you, and what are you allowed to access?”

PAM answers: “You need privileged access—how do we control, approve, limit, monitor, and expire it?”

In this project:

- **IAM:** Okta → IAM Identity Center → Permission Set → IAM Role → temporary AWS access.
- **PAM:** Access request → approval → JIT privilege → Vault/credential controls → controlled access → audit.